

Report Exercise 3
Advanced Computer Architectures

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Class: Master Software Engineering

1 Introduction

In the third lab of advanced computer architecture, we learned about the impact of 'warping,' 'thread divergence,' and 'striding' on the computation time of the GPU. We achieved this by conducting various tests on a kernel that inverts an image for each color component. This report presents our findings.

2 Implementation

For the implementation of the kernel that inverts an image, we utilized 'striding.' We opted for striding because an image (converted to a 1D array) often contains a vast number of elements. In each iteration, we incremented the index of a thread by $numberofblocks * blocksize$. This way, all elements of the array were covered.
In the second part of the lab, we had to make a small adjustment to our kernel. The code remained largely the same; only, for each R-element, a computation with higher complexity had to be performed. This introduces the concept of 'thread divergence' in our kernel, meaning that the threads of a warp responsible for calculating the R-element will take longer than the rest of the warp.

3 Evaluation & Conclusion

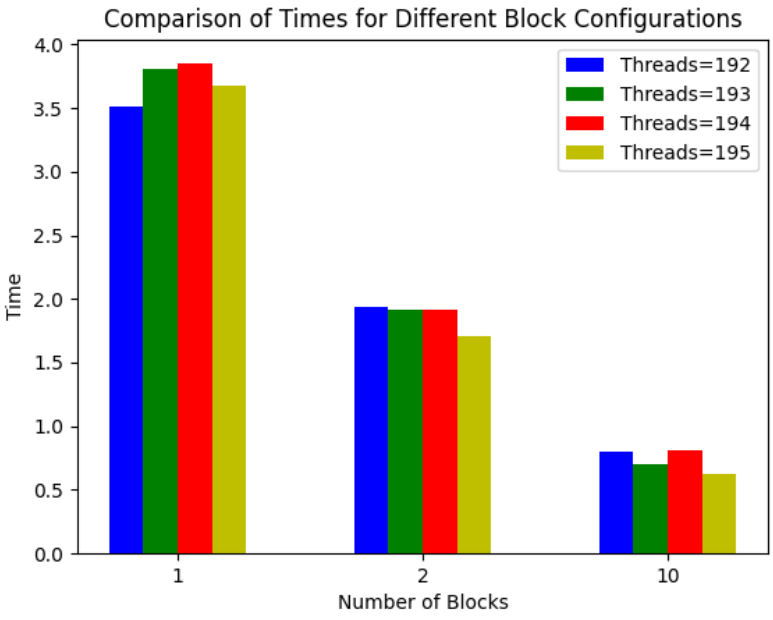


Figure 1: Your Caption Here

4 Evaluation & Conclusion

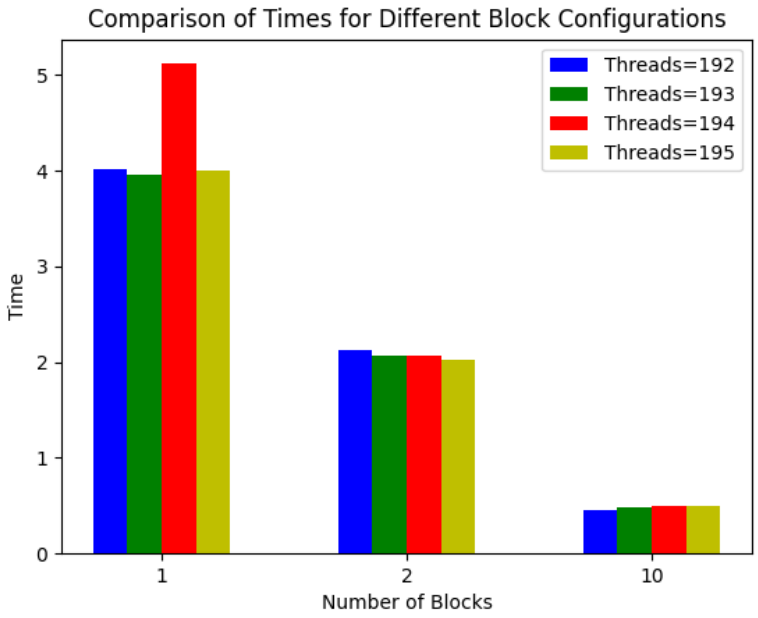


Figure 2: Your Caption Here

The following table 3 shows, for the simplest test cases run, the consequence of the ThreadsPerBlock choice. A very small reduction in number of iterations (due to the extra thread) results in vastly more reserved cores within the Stream Multiprocessor. This is because an SM iterates a full warp each time, even if not all 32 threads are used. [1] In the barcharts, we see the processing time at different choices of threads/block. We can best see the effect of a bad choice at 1 block by looking at the difference at 192 and 193 threads/block. When using more blocks (and therefore more threads in total), the phenomenon becomes harder to observe because the total execution time is much lower.

#Blocks	#Threads/B	#Warps	#Cores	#Iterations
1	192	6	192	10351
1	193	7	224	10298

Figure 3: Threads Per Block influence

[1] image array size: 1,987,392 elements